

Control sensitivity will differ amongst various airplanes. Because of the higher speeds flown, the control surfaces are more effective and a variation of just a few degrees in pitch attitude in a jet can result in over twice the rate of altitude change that would be experienced in a slower airplane. The sensitive pitch control in jet airplanes is one of the first flight differences that the pilot may notice, and the transitioning pilot may have a tendency to overcontrol pitch during initial training flights. Accurate and smooth control is one of the first techniques the transitioning pilot should master. Rather than gripping the yoke with the hand at high speeds, just using fingertips will result in smoother control inputs.

The pilot flying a swept wing jet airplane should understand that it is normal to fly at higher angles of attack. Depending on weight, density altitude, and available thrust, the pitch angle on takeoff may seem high. It is also not unusual to have a noticeable nose-up pitch on an approach to a landing.

Jet Airplane Takeoff and Climb

The following information is generic in nature and, since most civilian jet airplanes require a minimum flight crew of two pilots, assumes a two-pilot crew. If any of the following information conflicts with FAA-approved AFM procedures for a particular airplane, the AFM procedures take precedence. Also, if any of the following procedures differ from the FAA-approved procedures developed for use by a specific air operator and/or for use in an FAA-approved training center or pilot school curriculum, the FAA-approved procedures for that operator and/or training center/pilot school take precedence.

V-Speeds

The following are speeds that affect the jet airplane's takeoff performance. The jet airplane pilot should understand how to use these speeds when planning for takeoff.

- V_S —stalling speed or minimum steady flight speed at which the airplane is controllable.
- V_1 —critical engine failure speed or takeoff decision speed. It is the speed at which the pilot is to continue the takeoff in the event of an engine failure or other serious emergency. At speeds less than V_1 , it is considered safer to stop the aircraft within the accelerate-stop distance. It is also the minimum speed in the takeoff, following a failure of the critical engine at VEF, at which the pilot can continue the takeoff and achieve the required height above the takeoff surface within the takeoff distance.
- V_{EF} —speed used during certification at which the critical engine is assumed to fail.
- V_R —rotation speed, or speed at which the rotation of the airplane is initiated to takeoff attitude. This speed cannot be less than V_1 or less than $1.05 \times V_{MCA}$ (minimum control speed in the air). On a single-engine takeoff, it also allows for the acceleration to V_2 at the 35-foot height at the end of the runway.
- V_{LOF} —lift-off speed, or speed at which the airplane first becomes airborne. This is an engineering term used when the airplane is certificated to meet certain requirements. The pilot takes this speed into consideration if the AFM lists it.
- V_2 —takeoff safety speed, or a referenced airspeed obtained after lift-off at which the required one-engine-inoperative climb performance can be achieved.

Takeoff Roll

After confirming the runway and position match expectations, the airplane should be aligned in the center of the runway. When runway length is limited, the brakes should be held while the thrust levers are brought to a power setting specified in the AFM and the engines allowed to stabilize. The engine instruments should be checked for proper operation before the brakes are released or the power increased further. This procedure assures symmetrical thrust during the takeoff roll and aids in prevention of overshooting the desired takeoff thrust setting. After brake release, the power levers should be set to the pre-computed takeoff power setting and takeoff thrust adjustments made prior to reaching 60 knots. The final engine power adjustments are normally made by the pilot not flying. Retarding a thrust lever would only be necessary in case an engine exceeds any limitation.

Takeoff data, including V_1/V_R and V_2 speeds, takeoff power settings, and required field length should be computed prior to each takeoff. For any make and model without an FMS, the data should be recorded on a takeoff data card. This data is based on airplane weight, runway length available, runway gradient, field temperature, field barometric pressure, wind, icing conditions, and runway condition. Both pilots should review the takeoff data entered in an FMS or separately compute the takeoff data and cross-check with the takeoff data card. If takeoff plans change while taxiing, the pilot or crew should recalculate the takeoff data.

Captain's Briefing

I will advance the thrust levers.

Follow me through on the thrust levers.

Monitor all instruments and warning lights on the takeoff roll and call out any discrepancies or malfunctions observed prior to V_1 , and I will abort the takeoff. Stand by to arm thrust reversers on my command.

Give me a visual and oral signal for the following:

- 80 knots, and I will disengage nosewheel steering.
- V_1 , and I will move my hand from thrust to yoke.
- V_R , and I will rotate.

In the event of engine failure at or after V_1 , I will continue the takeoff roll to V_R , rotate and establish V_2 climb speed. I will identify the inoperative engine, and we will both verify. I will accomplish the shutdown, or have you do it on my command.

I will expect you to stand by on the appropriate emergency checklist.

I will give you a visual and oral signal for gear retraction and for power settings after the takeoff.

Our VFR emergency procedure is to.....

Our IFR emergency procedure is to.....

Figure 16-12. Sample captain's briefing.

A captain's briefing is an essential part of crew resource management (CRM) procedures and should be accomplished prior to takeoff. [Figure 16-12]

If sufficient runway length is available, a “rolling” takeoff may be made without stopping at the end of the runway. Using this procedure, as the airplane rolls onto the runway, the thrust levers should be smoothly advanced to the recommended intermediate power setting and the engines allowed to stabilize, and then proceed as in the static takeoff outlined above. Rolling takeoffs can also be made from the end of the runway by advancing the thrust levers from idle as the brakes are released.

During the takeoff roll, the pilot flying should concentrate on directional control of the airplane. This is made somewhat easier because there is no torque-produced yawing in a jet as there is in a propeller-driven airplane. The airplane should be maintained exactly on centerline with the wings level. This automatically aids the pilot when contending with an engine failure. If a crosswind exists, the wings should be kept level by displacing the control wheel into the crosswind. During the takeoff roll, the primary responsibility of the pilot not flying is to closely monitor the aircraft systems and to call out the proper V speeds as directed in the captain’s briefing.

Slight forward pressure should be held on the control column to keep the nose-wheel rolling firmly on the runway. If nose-wheel steering is being utilized, the pilot flying should monitor the nose-wheel steering to about 80 knots (or V_{MCG} for the particular airplane) while the pilot not flying applies the forward pressure. After reaching V_{MCG} , the pilot flying should bring his or her left hand up to the control wheel. The pilot’s other hand should be on the thrust levers until at least V_1 speed is attained. Although the pilot not flying maintains a check on the engine instruments throughout the takeoff roll, the pilot flying (pilot-in-command) makes the decision to continue or reject a takeoff for any reason. A decision to reject a takeoff requires immediate retarding of thrust levers.

The takeoff and climb-out should be accomplished in accordance with a standard takeoff and departure profile developed for the particular make and model airplane. [Figure 16-13]

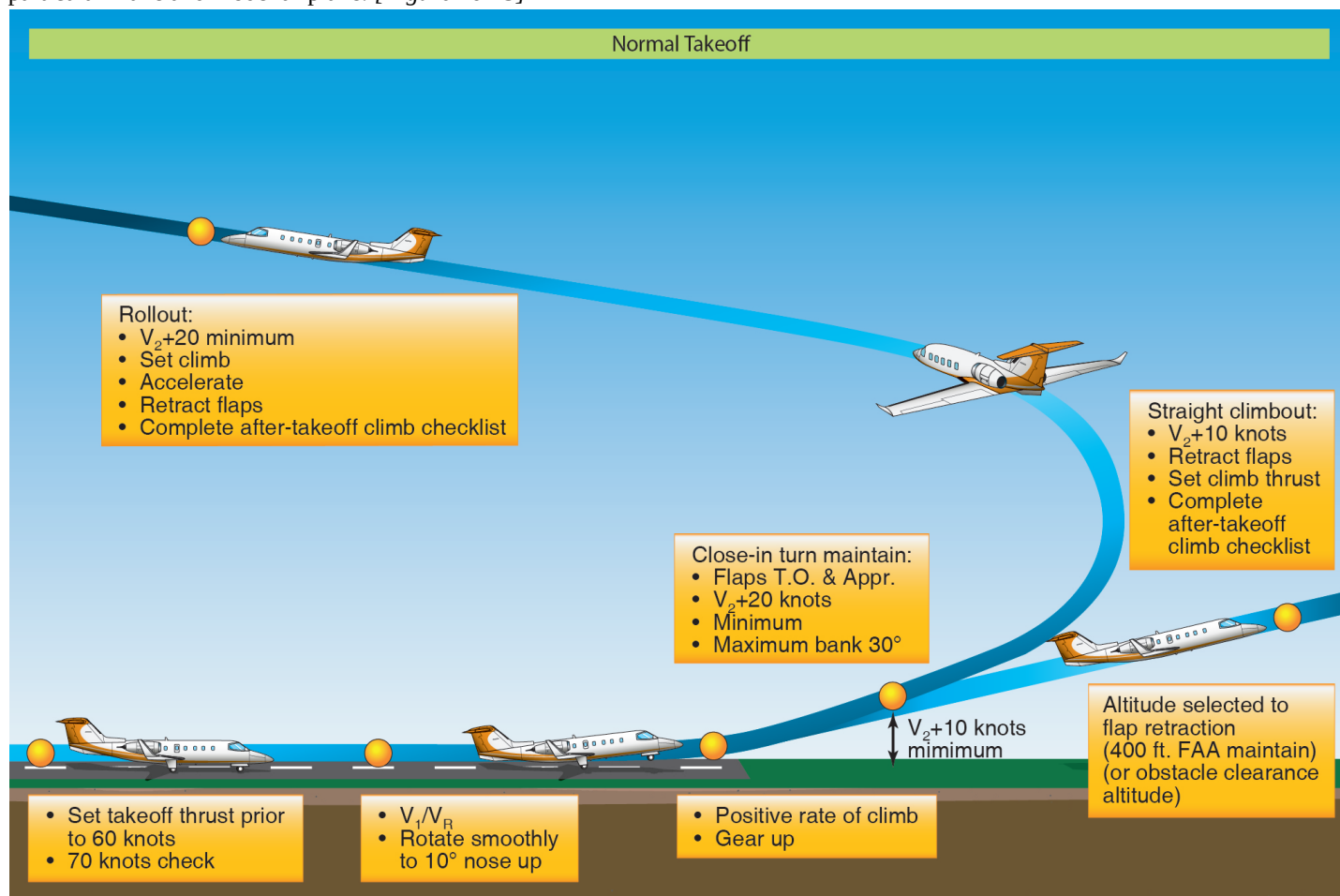


Figure 16-13. Sample takeoff and departure profile.

The pilot not flying should call out V_1 . After passing V_1 speed on the takeoff roll, it is no longer mandatory for the pilot flying to keep a hand on the thrust levers. The point for abort has passed, and both hands may be placed on the control wheel. As the airspeed approaches V_R , the control column should be moved to a neutral position. As the pre-computed V_R speed is attained, the pilot not flying should make the appropriate call-out, and the pilot flying should smoothly rotate the airplane to the appropriate takeoff pitch attitude.

Rejected Takeoff

Every takeoff could potentially result in a rejected takeoff (RTO) for a variety of reasons: engine failure, fire or smoke, unsuspected equipment on the runway, bird strike, blown tires, direct instructions from the governing ATC authority, or recognition of a significant abnormality (split-airspeed indications, activation of a warning horn, etc.).

Ill-advised rejected takeoff decisions by flight crews and improper pilot technique during the execution of a rejected takeoff contribute to a majority of takeoff-related commercial aviation accidents worldwide. Statistically, although only 2 percent of rejected takeoffs are in this category, high-speed aborts above 120 knots account for the vast majority of RTO overrun accidents. A brief moment of indecision may mean the difference between running out of runway and coming to a safe halt after an aborted takeoff.

It is paramount to remember that FAA-approved takeoff data for any aircraft is based on aircraft performance demonstrated in ideal conditions, using a clean, dry runway, and maximum braking (reverse thrust is not used to compute stopping distance). In reality, stopping performance can be degraded by an array of factors as diversified as:

- Reduced runway friction (grooved/non-grooved)
- Mechanical runway contaminants (rubber, oily residue, debris)
- Natural contaminants (standing water, snow, slush, ice, dust)
- Wind direction and velocity
- Low air density
- Flap configuration
- Bleed air configuration
- Underinflated or failing tires
- Penalizing MEL or CDL items
- Deficient wheel brakes or RTO auto-brakes
- Inoperative anti-skid
- Pilot technique and individual proficiency

Taking pilot response times into account, the go/no-go decision should be made before V_1 so that deceleration can begin no later than V_1 . If braking has not begun by V_1 , the decision to continue the takeoff is made by default. Delaying the RTO maneuver by just one second beyond V_1 increases the speed 4 to 6 knots on average. Knowing that crews require 3 to 7 seconds to identify an impending RTO and execute the maneuver, it stands to reason that a decision should be made prior to V_1 in order to ensure a successful outcome of the rejected takeoff. This prompted the FAA to expand on the regulatory definition of V_1 and to introduce a couple of new terms through the publication of Advisory Circular (AC) 120-62, "Takeoff Safety Training Aid."

The expanded definition of V_1 is as follows:

- a.) V_1 —the speed selected for each takeoff, based upon approved performance data and specified conditions, which represents:
 - 1.) The maximum speed by which a rejected takeoff assures that a safe stop can be completed within the remaining runway or runway and stopway;
 - 2.) The minimum speed which assures that a takeoff can be safely completed within the remaining runway, or runway and clearway, after failure of the most critical engine at the designated speed; and
 - 3.) The single speed which permits a successful stop or continued takeoff when operating at the minimum allowable field length for a particular weight.
- b.) Minimum V_1 —the minimum permissible V_1 speed for the reference conditions from which the takeoff can be safely completed from a given runway, or runway and clearway, after the critical engine had failed at the designated speed.
- c.) Maximum V_1 —the maximum possible V_1 speed for the reference conditions at which a rejected takeoff can be initiated and the airplane stopped within the remaining runway, or runway and stopway.
- d.) Reduced V_1 —a V_1 less than maximum V_1 or the normal V_1 , but more than the minimum V_1 , selected to reduce the RTO stopping distance required.

The main purpose for using a reduced V_1 is to properly adjust the RTO stopping distance in light of the degraded stopping capability associated with wet or contaminated runways, while adding approximately 2 seconds of recognition time for the crew.

Most aircraft manufacturers recommend that operators identify a “low-speed” regime (i.e., 80 knots and below) and a “high-speed” regime (i.e., 100 knots and above) of the takeoff run. In the “low-speed” regime, pilots should abort takeoff for any malfunction or abnormality (actual or suspected). In the “high-speed” regime, takeoff should only be rejected because of catastrophic malfunctions or life-threatening situations. Pilots should weigh the threat against the risk of overshooting the runway during an RTO maneuver. Standard operating procedures (SOPs) should be tailored to include a speed call-out during the transition from low-speed to high-speed regime, the timing of which serves to remind pilots of the impending critical window of decision-making, to provide them with a last opportunity to crosscheck their instruments, to verify their airspeed, and to confirm that adequate takeoff thrust is set, while at the same time performing a pilot incapacitation check through the “challenge and response” ritual.

Brakes provide the most effective stopping force, but experience has shown that the initial tendency of a flight crew is to use normal after-landing braking during a rejected takeoff. Delaying the intervention of the primary deceleration force during an RTO maneuver, when every second counts, increases stopping distance. Instead of braking after the throttles are retarded and the spoilers are deployed (normal landing), pilots should apply maximum braking immediately while simultaneously retarding the throttles, with spoiler extension and thrust reverser deployment following in short sequence. Differential braking applied to maintain directional control also diminishes the effectiveness of the brakes. A blown tire will eliminate any kind of braking action on that particular tire, and could also lead to the failure of adjacent tires.

In order to better assist flight crews in making a split-second go/no-go decision during a high-speed takeoff run, and avoid an unnecessary high-speed RTO, some commercial aircraft manufacturers have gone as far as inhibiting aural or visual malfunction warnings of non-critical equipment beyond a preset speed. The purpose is to prevent an overreaction by the crew and a tendency to select a risky high-speed RTO maneuver over a safer takeoff with a non-critical malfunction. Indeed, the successful outcome of a rejected takeoff, one that concludes without damage or injury, may be influenced by equipment characteristics.

In summary, a rejected takeoff should be perceived as an emergency. RTO safety could be vastly improved by:

- Developing SOPs aiming to advance the expanded FAA definitions of takeoff decision speed and their practical application, including the use of progressive callouts to identify transition from low-speed to high-speed regime.
- Promoting recognition of emergency versus abnormal situations through enhanced CRM training.
- Encouraging crews to carefully consider factors that may affect or even compromise available performance data.
- Expanding practical training in the proper use of brakes, throttles, spoilers, and reverse thrust during RTO demonstrations.
- Encouraging aircraft manufacturers to eliminate non-critical malfunction warnings during the takeoff roll at preset speeds.

Rotation and Lift-Off

Rotation and lift-off in a jet airplane requires planning, precision, and a fine control touch. The objective is to initiate the rotation to takeoff pitch attitude exactly at V_R so that the airplane accelerates through V_{LOF} and attains V_2 speed at 35 feet AGL. Rotation to the proper takeoff attitude too soon may extend the takeoff roll or cause an early lift-off, which results in a lower rate of climb and a divergence from the predicted flightpath. A late rotation, on the other hand, results in a longer takeoff roll, exceeding V_2 speed, and a takeoff and climb path below the predicted path.

Each airplane has its own specific takeoff pitch attitude that remains constant regardless of weight. The takeoff pitch attitude in a jet airplane is normally between 10° and 15° nose up. The rotation to takeoff pitch attitude should be made smoothly but deliberately and at a constant rate. Depending on the particular airplane, the pilot should plan on a rate of pitch attitude increase of approximately 2.5° to 3° per second.

In training, it is common for the pilot to overshoot V_R and then overshoot V_2 because the pilot not flying calls for rotation at or just past V_R . The pilot flying may visually verify V_R and then rotate late. If the airplane leaves the ground at or above V_2 , the excess airspeed may be of little concern on a normal takeoff. However, a delayed rotation can be critical when runway length or obstacle clearance is limited. On some airplanes, the rapidly increasing airspeed may cause the achieved flightpath to fall below the engine-out scheduled flightpath unless flying correct speeds. Rotation at the right speed and rate to the right attitude gets the airplane off the ground at the right speed and within the right distance.

Initial Climb

Once the proper pitch attitude is attained, the pilot should maintain it. Takeoff power is also maintained and the airspeed allowed to accelerate. Landing gear retraction should be accomplished after a positive rate of climb has been established and confirmed. In some airplanes gear retraction may temporarily increase the airplane drag while landing gear doors open. Premature gear retraction may cause the airplane to settle back toward the runway surface. In addition, the vertical speed indicator and the altimeter may not show a positive climb until the airplane is 35 to 50 feet above the runway due to ground effect.

The pilot should hold the climb pitch attitude as the airplane accelerates to flap retraction speed. However, the flaps should not be retracted until obstruction clearance altitude or 400 feet AGL has been passed. Ground effect and landing gear drag reduction result in rapid acceleration during this phase of the takeoff and climb. Airspeed, altitude, climb rate, attitude, and heading should be monitored carefully. As the airplane develops a steady climb, longitudinal stick forces can be trimmed out. If making a power reduction, the pilot should reduce the pitch attitude simultaneously if needed and monitor the airplane airspeed and rate of climb so as to preclude an inadvertent reduction in desired performance or a descent.

Speed is limited to 250 KIAS below 10,000 feet MSL in the United States unless otherwise authorized by the Administrator (14 CFR part 91, section 91.117(a)). At or above that altitude, the best rate of climb speed is published in the AFM. If asked to increase rate of climb, increasing pitch slightly will have the desired effect as airspeed bleeds off. If the airplane slows to L/D_{MAX} , the airplane is at its best angle of climb speed, but the rate of climb is less than it was at best rate of climb speed. Trading airspeed for altitude and a temporary increased rate of climb is referred to as a “zoom climb.” This type of climb provides an increased rate of climb for a few thousand feet, but it ultimately reduces overall climb performance.

Jet Airplane Descent and Approach

The smoothest and most fuel-efficient descent would be to reduce power to flight idle and slow to L/D_{MAX} . In this scenario, the pilot would descend, level off to decelerate, configure for landing, intercept the final approach, and continue a gradual deceleration until setting power for a stabilized descent on final. Traffic and time considerations almost always require deviation from this example, and the typical descent profile has three descent segments with two speed reductions in between.

Descent Planning

For a typical idle power descent, the top of descent (TOD), point A in *figure 16-14*, is determined by altitude, adjusted for wind. Jet descent profiles normally approximate a 3 degree path, with some time/distance required for deceleration in level flight. While exact distances will vary, having a descent plan will put the pilot well ahead of the jet and in a better position to monitor the automation.

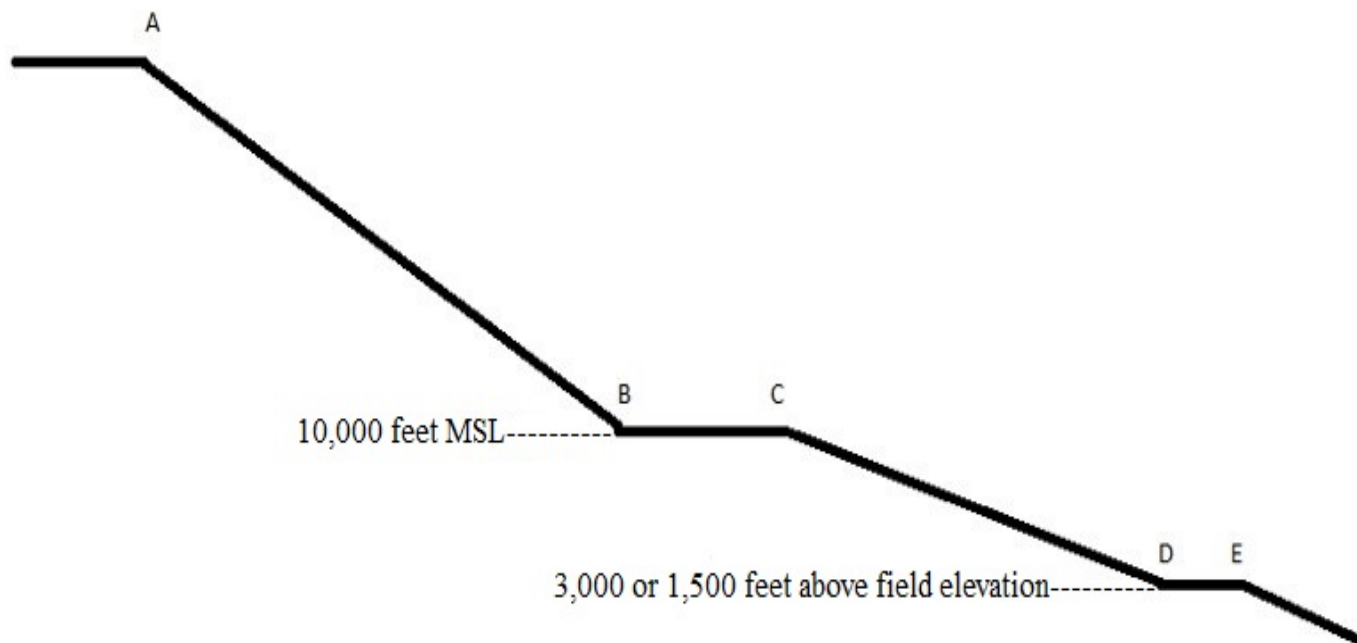


Figure 16-14. Typical descent profile.

For a straight-in VFR approach to an airport without factoring wind, an estimate for TOD may be calculated by multiplying the planned descent (in thousands of feet) by 3 and adding any distance needed for speed reductions in level flight (losing about 10 KIAS per mile when level). If flying at 35,000 feet above airport elevation, a cruise descent would start approximately 120 miles from the airport (35 times 3, plus about 15 miles for speed reduction, in stages, from cruise speed in this example). [Figure 16-14] Normally, cruise Mach is maintained until increasing air density causes indicated airspeed to increase to the desired descent speed, which usually occurs just below 30,000 feet. If arriving at point B at 10,000 feet MSL about 40 miles from the airport for deceleration to 250 knots, the pilot would resume a descent about 35 miles from the airport, continuing to 1,500 feet about 15 miles from the runway. The approach would continue with deceleration and flap extension so as to start the final descent 5 miles from the runway. There, the pilot extends the landing gear and selects landing flaps by 1,000 feet AGL, and brings the power up by 500 feet AGL to maintain the appropriate speed for a stabilized approach.

Variables that affect the TOD calculation include:

- Head/tail wind component (adjust distance 1 mile for each 10 knots of wind at cruise altitude),
- Field elevation,
- Terrain considerations,
- Runway alignment on arrival,
- ATC vectors and speed restrictions,
- Type of approach.

Descent Energy Management

While descending, the pilot can check the progress periodically. Estimating using round numbers keeps the calculation simple. Passing 25,000 feet should occur at 75 miles out plus or minus corrections; 20,000 feet should be at 60 miles, etc. If there is a deviation from the desired altitude/distance target, the energy state needs to be adjusted.

As discussed in Chapter 4, *Using Energy Management to Master Altitude and Airspeed Control*, there are two forms of energy in an airplane: potential energy in the form of altitude, and kinetic energy in the form of speed. In the normal operating regime at speeds above L/D_{MAX} , increasing speed increases total drag, while a decreasing speed will decrease total drag.

At idle power and at speeds above L/D_{MAX} , increasing speed increases the rate of descent. Sample data for a particular make and model might look like the following:

- 210 KIAS = 1,000 feet per minute
- 250 KIAS = 1,500 feet per minute
- 300 KIAS = 3,000 feet per minute

The exponential increase in parasite drag at higher speeds has a significant impact on both the rate of descent and the descent angle. Using the sample numbers, a 20% increase in airspeed from 210 to 250 knots, results in a 50% increase in the descent rate. However, a 20% increase in airspeed from 250 to 300 knots results in a 100% increase in the descent rate. Therefore, when at a higher altitude than desired in a descent, lowering the nose to increase speed will increase the descent angle and get the aircraft back to the desired path. Conversely, if lower than planned in descent, raising the nose to decrease speed will reduce descent angle until back on the desired path. Often, just a 10-knot change in speed allows for a smooth and gradual correction.

If speed adjustment is not an option, power can be added to correct a low-energy state, or the speed brakes used to correct a high-energy state. Numerous power fluctuations or repeated deployment and stowing of speed brakes is an indication of either pilot failure to adequately plan and/or manage the descent, or a poorly designed arrival procedure.

If a different descent speed from that planned is used during a descent, an adjustment should be made to the top of descent point. If ahead of schedule, leaving cruise altitude sooner, setting flight idle, and descending at a slower speed will burn less fuel. Conversely, if running late and willing to burn some extra fuel, the pilot can leave cruise later and descend at a higher speed. In all cases, the pilot should check progress during the descent and continue to adjust as necessary.

Planned descent speed will affect the position of the planned top of descent point. [Figure 16-15] In this example, both jets fly past point X at the same cruise speed and altitude with plans to arrive at point Y at 10,000 feet and 250 knots. In both cases, the aircraft would then be in a position to set up a continued descent. The 250-knot descent requires a few miles for deceleration and gives a shallower descent path. The 300-knot descent allows staying at altitude longer, descending at a steeper angle, and then leveling off to slow to 250 knots. The jet that descended at 300 knots arrives first at point Y but burns more fuel. While not depicted, an inefficient descent plan would start the descent at point X, maintain 300 knots, and require power to maintain that airspeed on a shallow descent path.

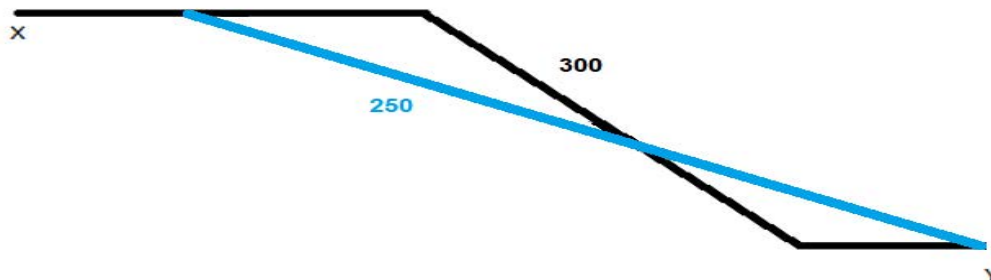


Figure 16-15. Effect of speed on descent path.

Descending prior to the planned TOD point will increase time to destination and fuel consumption. When given a descent clearance prior to the planned TOD, it is acceptable to ask ATC if the descent can be done at the pilot's discretion. If authorized to do so, this option allows for maintaining speed and altitude until reaching the calculated top of descent point. If an immediate descent is required, a descent at 1,000 feet per minute is usually acceptable until reaching the desired path. If a descent clearance has not been received by the planned TOD point, a speed reduction will reduce the airplane's kinetic and total energy while potential energy remains constant. When the clearance is received, a slightly steeper descent at the onset allows for a desired increase in kinetic energy at the expense of altitude and an appropriate descent rate such that the airplane follows the steeper desired path with acceptable energy distribution.

Jet Engine Landing

14 CFR part 25, section 25.125 defines the horizontal distance needed in order to land a jet airplane. The regulation describes the landing profile as the horizontal distance required to land and come to a complete stop from a point 50 feet above the landing surface. Manufacturers determine the landing distance on a dry, level runway at standard temperatures without using thrust reversers, auto brakes, or auto-land systems as a baseline. The pilot uses the landing weight and environmental conditions to determine the actual expected landing requirement based on the FAA-approved data in the AFM. As an accepted safety practice, pilots normally add a 40% cushion for landing on a dry runway. Dividing the usable runway length by 1.67 should give a number equal to or greater than the landing distance calculated from AFM data. For a wet runway, the distance should be increased by an additional 15%. [Figure 16-16]

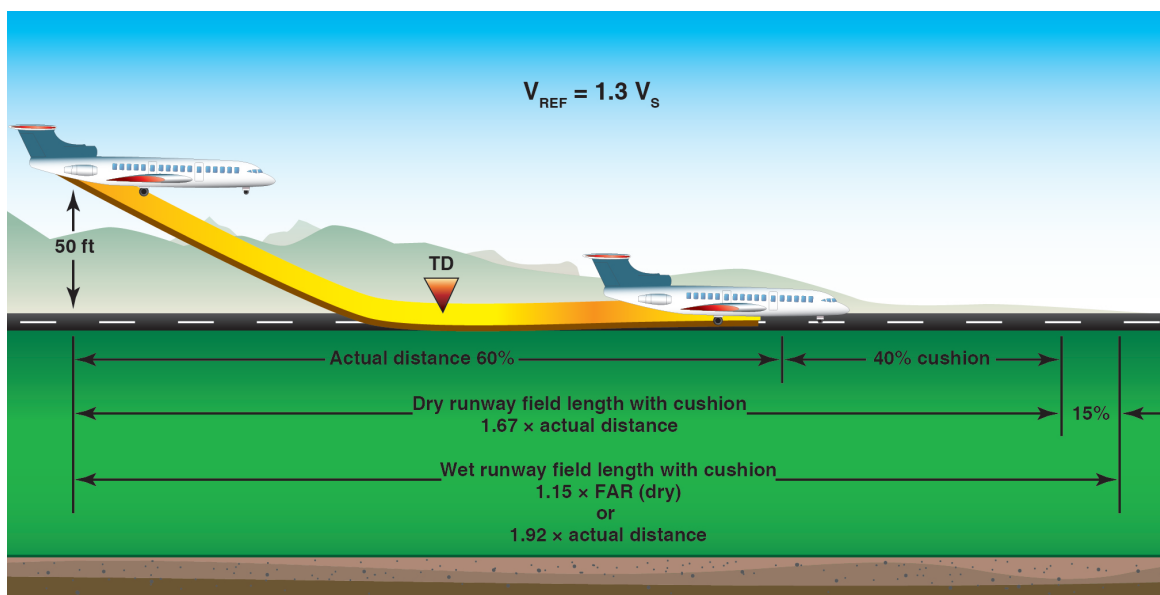


Figure 16-16. FAR landing field length required.

Simply put, the pilot divides the length of an intended runway by 1.67 or 1.92, as appropriate, to determine the minimum distance that should be available for landing. With this safety margin, it works out that the minimum dry runway field length should be at least 1.4 times the calculated air and ground distance needed, and the wet runway landing field length should be at least 1.61 times the calculated air and ground distance needed. Careful flight planning allows a pilot to determine how much load in terms of fuel, passengers, or cargo can be carried to a particular runway while still maintaining the desired safety margin. Depending on the destination, the load might need to be limited in order to protect the safety margin when landing. This is often complex, since fuel load has its own safety implications.

Certified landing field length requirements are computed for the stop made with speed brakes deployed and maximum wheel braking. Reverse thrust is not used in establishing the certified landing distances. However, reversers should definitely be used, if available.

Landing Speeds

As in the takeoff planning, there are certain speeds that should be taken into consideration when landing a jet airplane. The speeds are as follows:

- V_{SO} —stall speed in the landing configuration.
- V_{REF} —1.3 times the stall speed in the landing configuration.
- Approach climb—the speed that guarantees adequate performance in a go-around situation with an inoperative engine.
- Landing climb—the speed that guarantees adequate performance in arresting the descent and making a go-around from the final stages of landing with the airplane in the full landing configuration and maximum takeoff power available on all engines.

Pilots may need to perform traffic pattern takeoffs and landings. Pilots should use speeds recommended by the manufacturer while maneuvering in the traffic pattern prior to slowing to the final approach target speed in relation to V_{REF} . The speeds should be calculated for every landing and posted where they are visible to both pilots.

The approach and landing sequence in a jet airplane should be accomplished in accordance with an approach and landing profile developed for the particular airplane. [Figure 16-17]

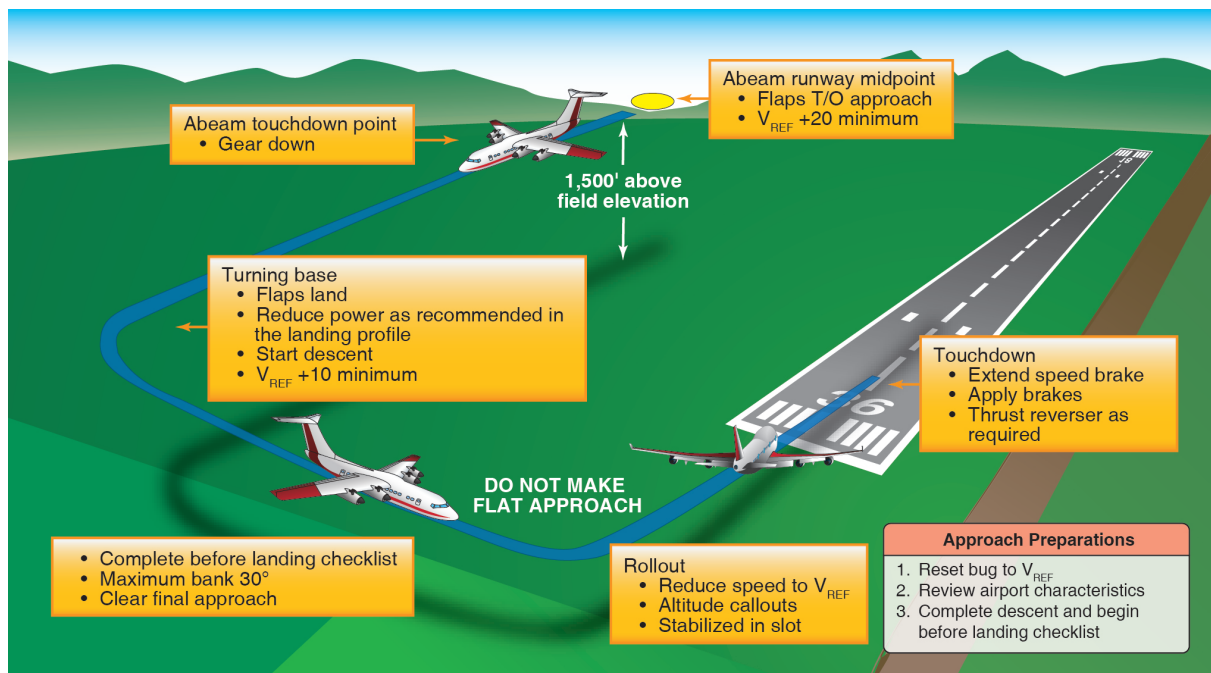


Figure 16-17. Typical approach and landing profile.

Significant Differences

A safe approach in any type of airplane culminates in a particular position, speed, and height over the runway threshold. That final flight condition is the target window at which the entire approach aims. Propeller-powered airplanes are able to approach that target from wider angles, greater speed differentials, and a larger variety of glidepath angles. Jet airplanes are not as responsive to power and course corrections, so the final approach should be more stable, more deliberate, and more constant in order to reach the window accurately.

The transitioning pilot should understand that in spite of their impressive performance capabilities, there are many reasons why jet airplanes are less forgiving than piston-engine airplanes during approaches and when correcting approach errors.

- There is no propeller slipstream to produce immediate extra lift at constant airspeed. There is no such thing as salvaging a misjudged glidepath with a sudden burst of power. Added lift can only be achieved by accelerating the airframe.
- Propeller slipstream is not available to lower the power-on stall speed. There is virtually no difference between power-on and power-off stall speed. It is not possible in a jet airplane to jam the thrust levers forward to avoid a stall.
- Jet engine response at low rpm is slower. This characteristic requires that the approach be flown at a stable speed and power setting on final so that sufficient power is available quickly if needed.
- Jet airplanes are consistently heavier and have faster approach speeds than a comparably sized propeller airplane. Since greater force is required to overcome momentum for speed changes or course corrections, the typical jet responds less quickly than the propeller airplane and requires careful planning and stable conditions throughout the approach.
- When the speed does increase or decrease, there is little tendency for the jet airplane to re-acquire the original speed. The pilot needs to make speed adjustments promptly in order to remain on speed.
- Drag increases faster than lift and produces a high sink rate at low speeds. Jet airplane wings typically have a large increase in drag in the approach configuration. When a sink rate does develop, the only immediate remedy is to increase pitch attitude (AOA). Because drag increases faster than lift, that pitch change rapidly contributes to an even greater sink rate unless a significant amount of power is promptly applied.

These flying characteristics of jet airplanes make a stabilized approach an absolute necessity.

Stabilized Approach

The performance charts and the limitations contained in the FAA-approved AFM are predicated on momentum values that result from programmed speeds and weights. Runway length limitations assume an exact 50-foot threshold height at an exact speed of 1.3 times V_{SO} . That “window” is critical and is a prime reason for the stabilized approach. Performance figures also assume that once through the target threshold window, the airplane touches down in a target touchdown zone approximately 1,000 feet down the runway, after which maximum stopping capability is used.

The basic elements to the stabilized approach are listed below as follows:

- The airplane should be in the landing configuration by 1,000 feet AGL in the approach. The landing gear should be down, landing flaps selected, trim set, and fuel balanced. Ensuring that these tasks are completed helps keep the number of variables to a minimum during the final approach.
- The airplane should be on profile before descending below 1,000 feet. Configuration, trim, speed, and glidepath should be at or near the optimum parameters early in the approach to avoid distractions and conflicts as the airplane nears the threshold window. An optimum glidepath angle of about 3° should be established and maintained.
- Indicated airspeed should be between zero and 10 knots above the target airspeed by 500 feet AGL. There are strong relationships between trim, speed, and power in most jet airplanes, and it is important to stabilize the speed in order to minimize those other variables.